

Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <signal.h>
4 #include <unistd.h>
5
6 void signal_handler(int sig)
7 {
8     if (sig == SIGINT)
9     {
10         printf("\nReceived SIGINT (Ctrl+C)\n");
11         printf("Interrupt signal handled.\n");
12         printf("Program terminated.\n");
13         exit(0);
14     }
15     else if (sig == SIGTERM)
16     {
17         printf("\nReceived SIGTERM\n");
18         printf("Termination signal handled.\n");
19         printf("Program terminated.\n");
20         exit(0);
21     }
22     else if (sig == SIGUSR1)
23     {
24         printf("\nReceived SIGUSR1\n");
25         printf("User-defined signal handled.\n");
26     }
27 }
28
29 int main()
30 {
31     // Register signal handlers
32     signal(SIGINT, signal_handler);
33     signal(SIGTERM, signal_handler);
34     signal(SIGUSR1, signal_handler);
35
36     printf("Signal handling program started.\n");
37     printf("Process ID: %d\n", getpid());
38     printf("Press Ctrl+C to generate SIGINT.\n");
39     printf("Use another terminal to send SIGUSR1 or SIGTERM.\n");
40
41     while (1)
42     {
43         printf("Program is running...\n");
44         sleep(3);
45     }
46
47     return 0;
48 }
49
```

Output:

```
saisurendra@SaiSurendra:~$ nano Practical22.c
saisurendra@SaiSurendra:~$ gcc Practical22.c -o Practical22
saisurendra@SaiSurendra:~$ ./Practical22
Signal handling program started.
Process ID: 548
Press Ctrl+C to generate SIGINT.
Use another terminal to send SIGUSR1 or SIGTERM.
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
```

```
Received SIGINT (Ctrl+C)
Interrupt signal handled.
Program terminated.
saisurendra@SaiSurendra:~$ kill -SIGUSR1 548
-bash: kill: (548) - No such process
```